

- 14** Batrachotoxin is a poison found in frogs in the Colombian jungle. The poison is used to produce poison darts.

The poison works by increasing the permeability of the cell surface membrane of nerve and muscle cells to sodium ions, which move out of the cells.

Four students made statements about how the poison affects the cells.

- 1 Water leaves the cells by osmosis, causing the cells to shrink.
- 2 Water enters the cells by osmosis, causing the cells to burst.
- 3 When the sodium ions move out of the cells the intracellular fluid has a more positive water potential than the extracellular fluid.
- 4 When the sodium ions move out of the cells the extracellular fluid has a more positive water potential than the intracellular fluid.

Which statements are correct for the cells affected by batrachotoxin?

- A** 1 and 3 **B** 1 and 4 **C** 2 and 3 **D** 2 and 4